NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

Data Science Fundamentals [SC3021]

Chapter 3: Data Science Ecosystems

Assoc. Prof. Melanie Herschel | CCDS

Course content

Data science overview

What is Data Science?

Data Science Thinking vs. Computational Thinking

Data Science Ecosystem

- **Ecosystem overview**
- **Component details**

Data preparation and data management

Data profiling

Structuring

Enriching

Data cleaning

Data management platforms

Schema design

Querying data

Data analytics and understanding

Descriptive analytics

Diagnostic Analytics

Prescriptive analytics

Predictive analytics

Data visualization

Design considerations

Privacy

Security

Ethics

Psychology

Policies

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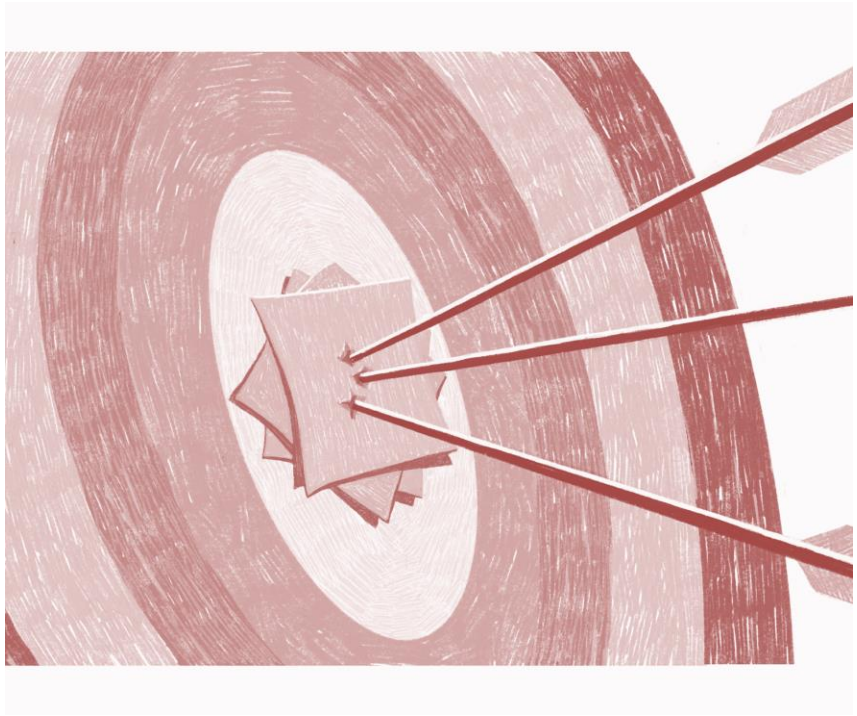
At the end of this lecture, you should be able to…

- … describe the main components of the data science ecosystem.
- … explain how different factors influence the development of a data science system.
- … define the functionality of data science building blocks at a high level to solve data science applications.
- … reflect on possible data protection, data ethics, social and policy aspects that may affect the data science application deployment.



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ECOSYSTEM OVERVIEW

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Applications, systems, and deployments

Data Science application	Problem instance that applies Data Science Thinking to solve a problem based on data
Data Science system	Software system that supports the process underlying Data Science Thinking
Data Science application deployment	Use of a data science system for a data science problem in a production environment

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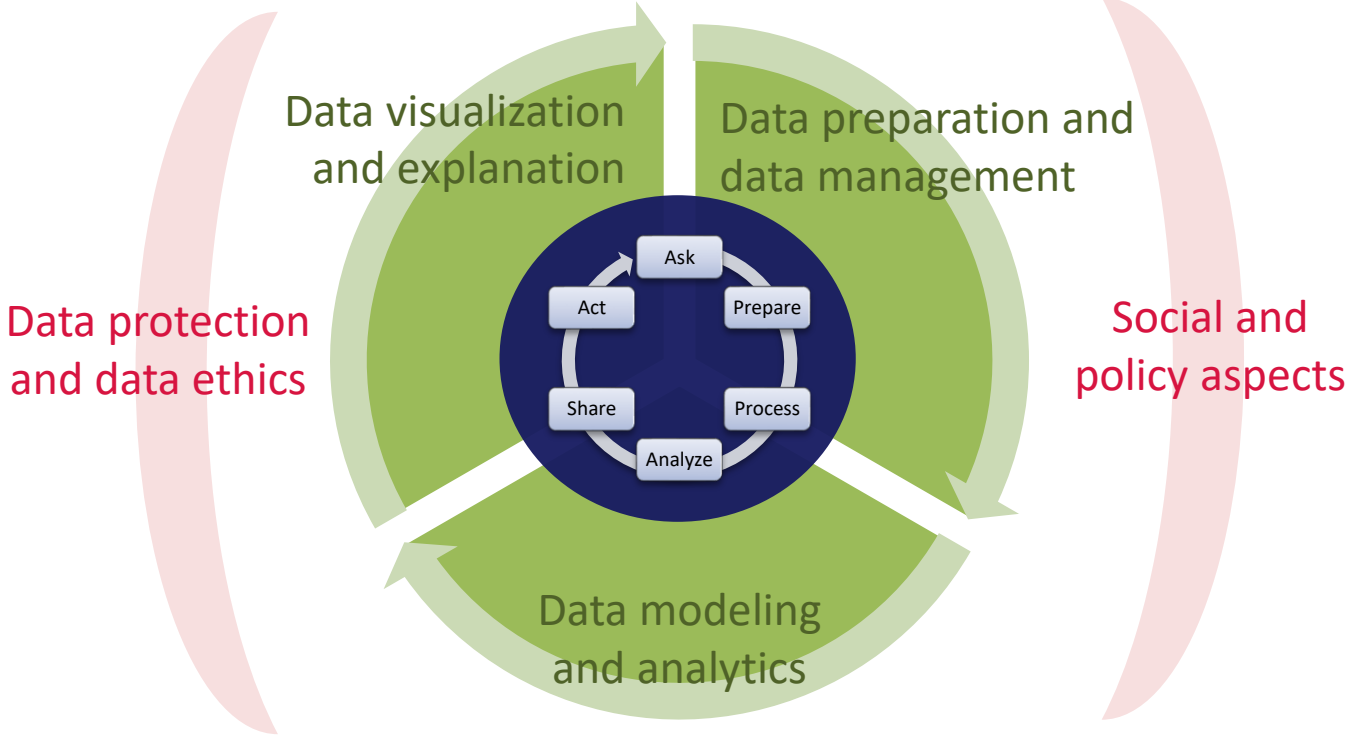
Applications, systems, and deployments

Data Science application

Data Science system

Data Science application deployment



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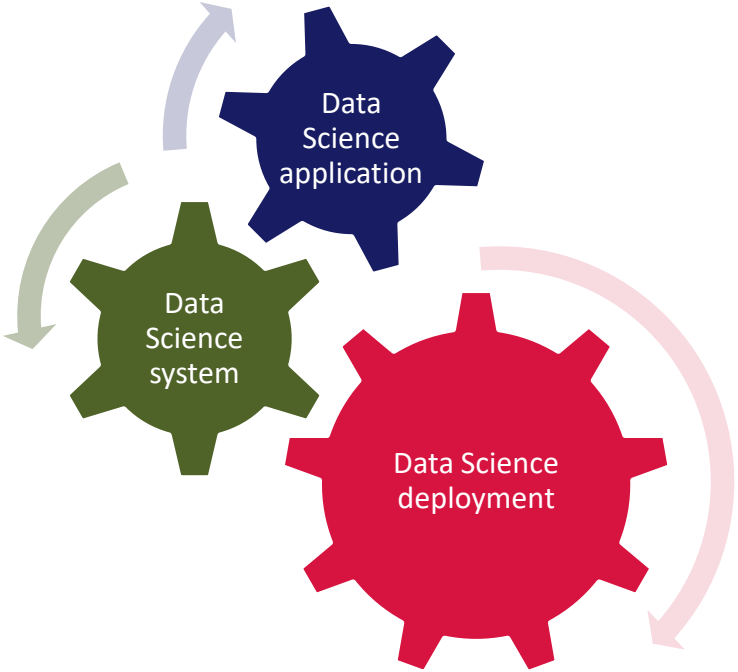
Holistic consideration



- Applications, systems, and deployments are interlinked and influence one another.
- Examples:
  - Data science systems provide tools and technologies that help data scientists address their problems.
  - Data science systems need to be carefully designed to fit the target applications.
  - Data science systems need to support deployments within their social and policy context.

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Data Science ecosystem

- A **framework** designed to guide the development of **data science systems**.
- The ecosystem is composed of **building blocks** that are application-informed and context-informed.
- **Application-informed**: Data Science applications guide the appropriate technologies, tools, algorithms, and methodologies to be incorporated in data science systems and leverage them to address the application needs.
- **Context-informed**: Social and policy context significantly impact data-science application deployments. Core technologies developed and used in data science systems need to take these into account and reflect them in application deployments.

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Illustration of application-informed

- **Application problem**: Classify if a product sold online will be returned or not.
- Some considerations for **system development** (there are more):
  - The application defines a **classification** problem.
  - The product description on the Web is semi-structured, so **restructuring** and **cleaning** of the data is necessary to obtain dense feature vectors.
  - The system needs to implement **standardization** and **alignment** to ensure that feature representations of online products are compatible with features of the available product transaction data.
  - The choice of **model** should ensure highly **accurate** classification results.

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Illustration of context-informed

- **Application problem**: Classify if a person applying for a loan will repay the loan or not.
- Some considerations for **system development** (there are more):
  - The application defines a **classification** problem.
  - Customer profiles vary and are semi-structured data, so **restructuring** and **cleaning** of the data is necessary to obtain dense feature vectors.
  - The system needs to implement **standardization** and **alignment** to ensure that feature representations of applicants are compatible with features of the available bank transaction data.
  - ~~The choice of **model** should ensure highly **accurate** classification results.~~
  - Choose a right **model** to obtain **accurate and fair** classification results.
  - Handling of personal data requires the integration of **privacy** mechanisms into the data science system.
  - Financial data needs to be protected through appropriate **security** mechanisms.

Changed social and policy context!

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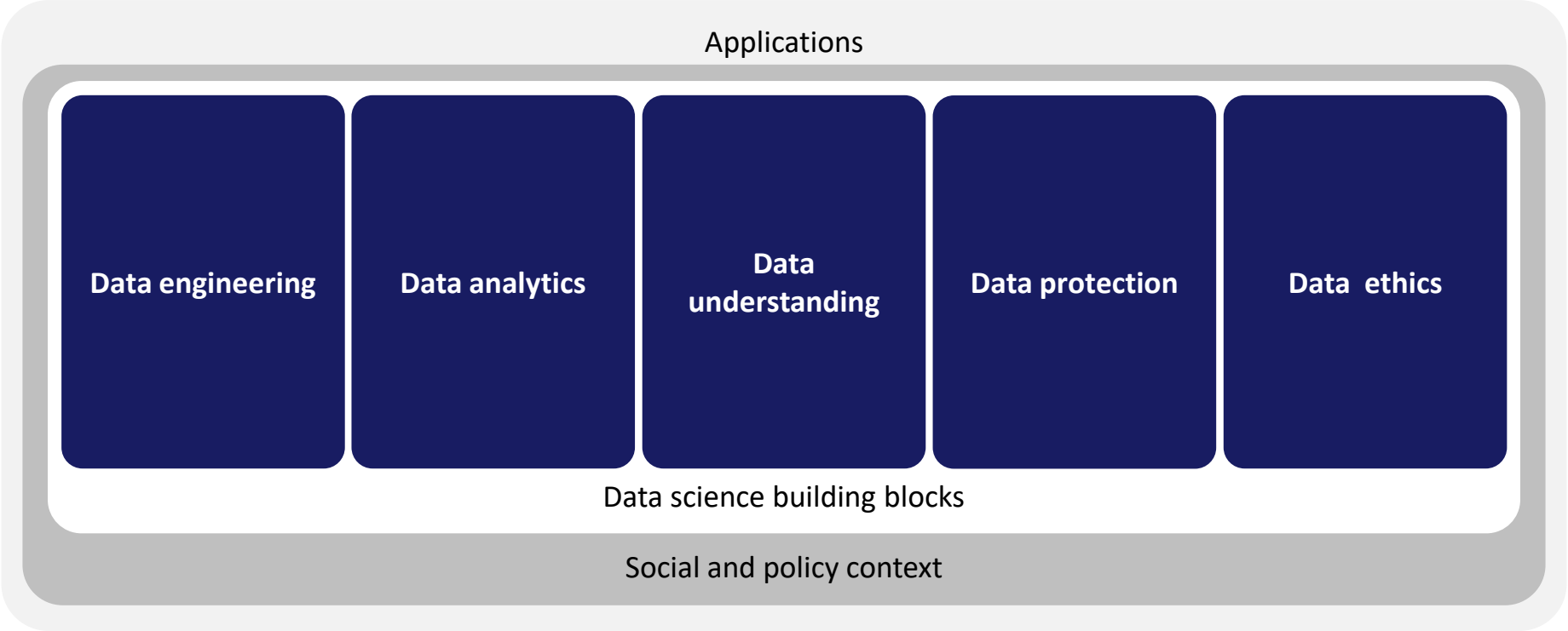
Data Science ecosystem



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Data Science ecosystem

Applications

Data engineering

- Data preparation
- Data management

Data analytics

- Data exploration (data mining)
- Models and algorithms (machine learning)

Data understanding

- Explanation
- Visualization

Data protection

- Security
- Privacy

Data ethics

- Impact on individuals, organizations, and society
- Bias in data
- Algorithmic bias
- Regulatory issues

Data science building blocks

Social and policy context

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Ecosystem structures the remainder of this lecture

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COMPONENT DETAILS

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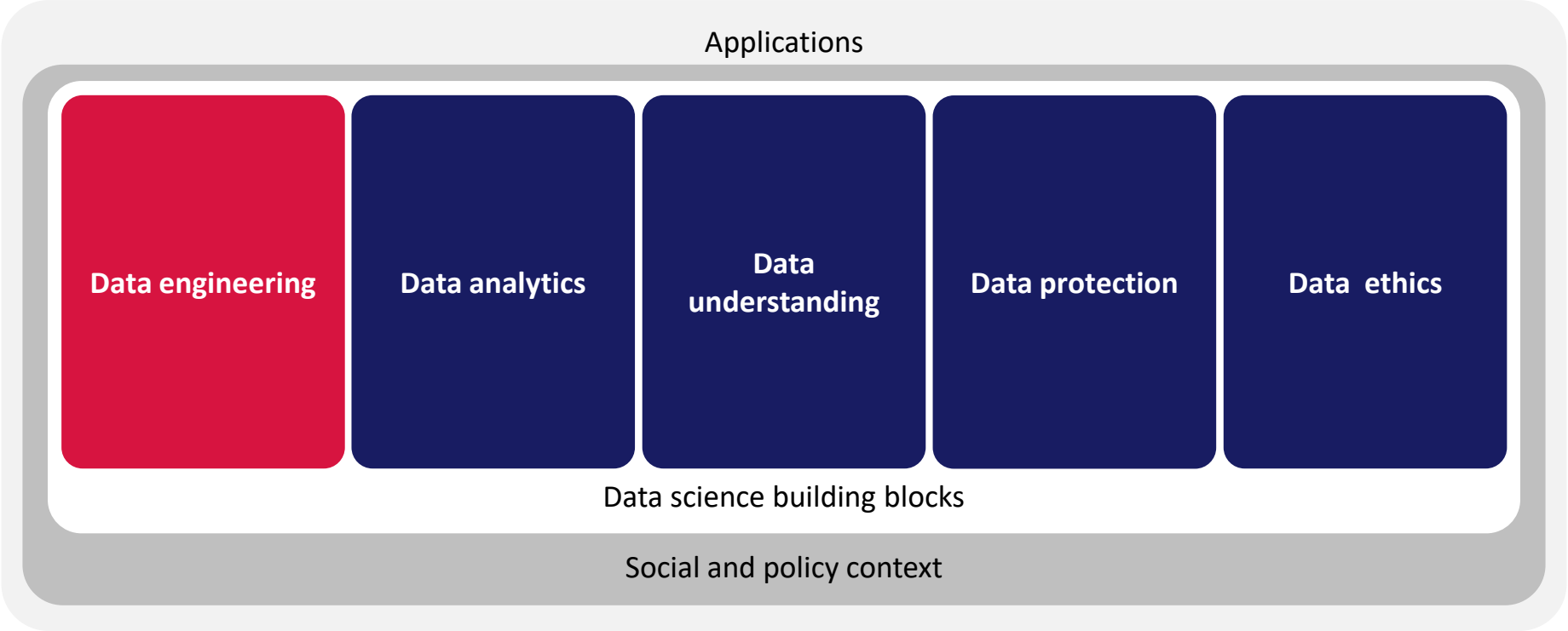
Data Science ecosystem



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Data Engineering

Data Engineering

The practice of designing, building, and maintaining systems that collect, manage, and convert raw data into usable information for downstream applications.

Data preparation

Operations typically involved in a data engineering process that transform the data.

Data management

Solutions devised to organize, protect, store, access, and manipulate data.

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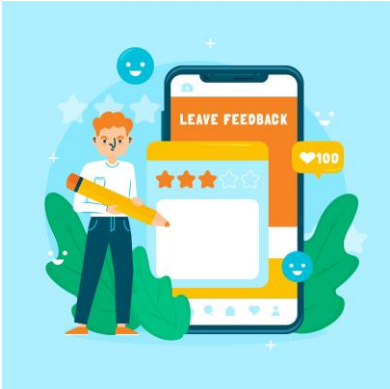
Data preparation

Operations transforming data in multiple **stages**:

- Data profiling
- Data structuring
- Data enriching
- Data cleaning



Running **example:**

- Goal: analyze the relationship between product ratings and customer demographics (age, gender, place of residence).
- Requires: data on both product reviews by customers and customer demographics.

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Data preparation - Example

Sample data from user_reviews

Product	Reviewer	Rating	Text	Date	City
iPhone 6	helloKitty93	4	…	19/12/2011	LA
Apple iPhone 6	⊥	3	…	04/24/2012	New York
Samsung Galaxy	SecretUser007	5	⊥	21/01/2018	Austin
⊥	helloKitty93	4	…	29/09/2020	CA

Sample data for customer_demographics

Username	Name	Age	isMale	isFemale	City
helloKitty	Jane Smith	46	⊥	1	Los Angeles
JD	John Doe	24	1	0	New York
SecretUser007	James Bond	65	1	0	⊥
maxMiller	Max Miller	37	⊥	⊥	Miami

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Data preparation - Profiling

- Examine and analyze data to create metadata (data about data).
- Metadata represent a summary of various characteristics of the data.
- These summaries help to assess usefulness and quality of data.

Statistics commonly used in summaries:

- Number of rows
- Min and max values
- Number of missing values
- Outliers
- Data type
- Data format violations
- Histograms
- …

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Data preparation - Profiling

Metadata for user_reviews

Column	Type	Missing values	Unique values
Product	String	55	10987
Reviewer	String	126	8426
Rating	Int	0	5
Text	String	6024	4963
Date	Date	0	365
City	String	3024	430

Metadata for customer_demographics

Column	Type	Missing values	Unique values
username	String	0	5000
Name	String	0	4230
Age	Int	10	89
isMale	Bool	2040	2
isFemale	Bool	3022	2
City	String	75	500

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Data preparation - Structuring

- Organize and transform the data into a well-defined format.
- The format should be suitable for the intended analysis.
- Properly structured data enables efficient and effective data processing.

Typical structuring tasks:

- Organization of data into tables.
- Table normalization to avoid data redundancy and improve data integrity.
- Splitting values
- Pivoting data
- Group numerical data into ranges
- …

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Data preparation - Structuring

Sample data from user_reviews

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Samsung Galaxy	SecretUser007	5	⊥	21/01/2018	Austin
⊥	helloKitty93	4	…	29/09/2020	CA

Restructured data

Product	Reviewer	Rating	Day	Month	Year	City
iPhone 6	helloKitty93	4	19	12	2011	LA
Apple iPhone 6	⊥	3	04	24	2012	New York
Samsung Galaxy	SecretUser007	5	21	01	2018	Austin
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Data preparation - Structuring

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maxMiller	Max Miller	37	⊥	⊥	Miami

Restructured data

Username	Name	Age	AgeRange	Gender	City
helloKitty	Jane Smith	46	46-60	F	Los Angeles
JD	John Doe	24	18-30	M	New York
SecretUser007	James Bond	65	60 - 75	M	⊥
maxMiller	Max Miller	37	30-45	⊥	Miami

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Data preparation - Enriching

- Enhance existing data by adding new and supplemental information.
- The supplemental information should be relevant to the data science application.
- Enrichment supports improving the accuracy, reliability, and completeness of the data for the task at hand.

Common enrichment steps:

- Add new attributes (columns)
- Join with external data sources
- Create derived attributes
- Data integration
- …

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Data preparation - Enriching

user_reviews (restructured)

Product	Reviewer	Rating	Day	Month	Year	City
iPhone 6	helloKitty93	4	19	12	2011	LA
Apple iPhone 6	⊥	3	04	24	2012	New York
Samsung Galaxy	SecretUser007	5	21	01	2018	Austin
⊥	helloKitty93	4	29	09	2020	CA

user_reviews (enriched)

Product	Category	Reviewer	Rating	Day	Month	Year	City	State
iPhone 6	Electronics	helloKitty93	4	19	12	2011	LA	CA
Apple iPhone 6	Electronics	⊥	3	04	24	2012	New York	NY
Samsung Galaxy	Electronics	SecretUser007	5	21	01	2018	Austin	TX
⊥	⊥	helloKitty93	4	29	09	2020	CA	⊥

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Data preparation - Enriching

Product	Category	Reviewer	Rating	Day	Month	Year	City	State
iPhone 6	Electronics	helloKitty93	4	19	12	2011	LA	CA
Apple iPhone 6	Electronics	⊥	3	04	24	2012	New York	NY
Samsung Galaxy	Electronics	SecretUser007	5	21	01	2018	Austin	TX
⊥	⊥	helloKitty93	4	29	09	2020	CA	⊥

Username	Name	Age	AgeRange	Gender	City
helloKitty	Jane Smith	46	46-60	F	Los Angeles
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SecretUser007	James Bond	65	60 - 75	M	⊥
maxMiller	Max Miller	37	30-45	⊥	Miami

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Data preparation - Enriching

user_reviews(enriched)

Product	Category	UN1	R	D	M	Y	City	State	UN2	Name	Age	Range	Gender	City2
iPhone 6	Electronics	helloKitty93	4	19	12	2011	LA	CA	helloKitty	Jane Smith	46	46-60	F	Los Angeles
Apple iPhone 6	Electronics	⊥	3	04	24	2012	New York	NY	⊥	⊥	⊥	⊥	⊥	⊥
Samsung Galaxy	Electronics	SecretUser007	5	21	01	2018	Austin	TX	SecretUser007	James Bond	65	60-75	M	⊥
⊥	⊥	helloKitty93	4	29	09	2020	CA	⊥	⊥	⊥	⊥	⊥	⊥	⊥

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Data preparation - Cleaning

- Identify and correct data errors, inconsistencies, and inaccuracies.
- Final step of data preparation that yields accurate, consistent, and reliable data suitable for analysis and decision-making.

Example data cleaning tasks:

- Identify matches
- Remove duplicates
- Remove irrelevant data
- Handle missing data
- Deal with outliers
- Make values consistent
- …

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Data preparation - Cleaning

user_reviews(cleaned)

Product	Category	UN1	R	D	M	Y	City	State	UN2	Name	Age	Range	Gender	City2
Apple iPhone 6	Electronics	helloKitty93	4	19	12	2011	Los Angeles	CA	helloKitty	Jane Smith	46	46-60	F	Los Angeles
Apple iPhone 6	Electronics	⊥	3	04	24	2012	New York	NY		⊥	⊥	⊥	⊥	New York
Samsung Galaxy A8	Electronics	SecretUser007	5	21	01	2018	Austin	TX	SecretUser007	James Bond	65	60-75	M	Austin
⊥	⊥	helloKitty93	4	29	09	2020	Los Angeles	CA	helloKitty	Jane Smith	46	46-60	F	Los Angeles

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Data preparation - Cleaning

user_reviews(cleaned)

Product	Category	R	D	M	Y	City	State	Name	Age	Range	Gender
Apple iPhone 6	Electronics	4	19	12	2011	Los Angeles	CA	Jane Smith	46	46-60	F
Samsung Galaxy A8	Electronics	5	21	01	2018	Austin	TX	James Bond	65	60-75	M

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Data preparation – Wrap-up

- Data preparation, through its different **stages**, transforms raw data **into high-quality, well-structured data** that is **suitable and ready-to-use** for the analysis needed by the target data science application.
- Such data are **imperative for meaningful analytical results** (prevents “garbage-in-garbage-out” problem).
- Although presented as subsequent stages, the **order may vary**, and some stages may apply more than once before producing the final dataset.

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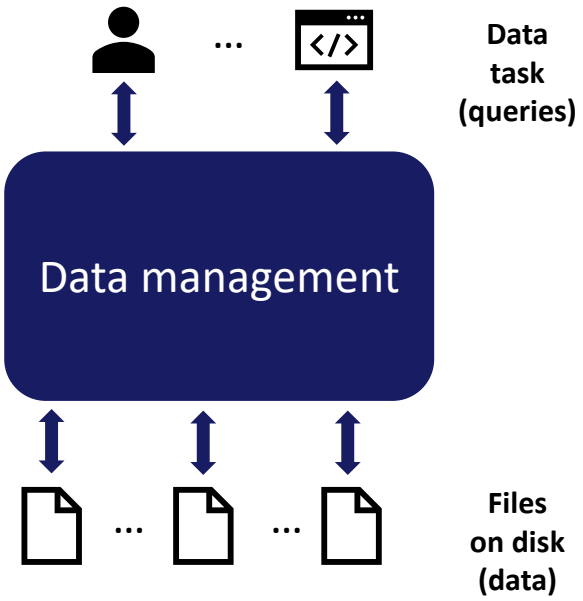
Data management

- Solutions devised for **easy support of operations on data**, e.g., organize, protect, store, access, and manipulate data.
- Systems, algorithms, and methods that support **efficient** and **scalable** use of data.
- Ease of use via **abstraction** of data handling issues from physical data storage on disc to processing user queries.
- **Many different types of data supported** by different data management solutions.



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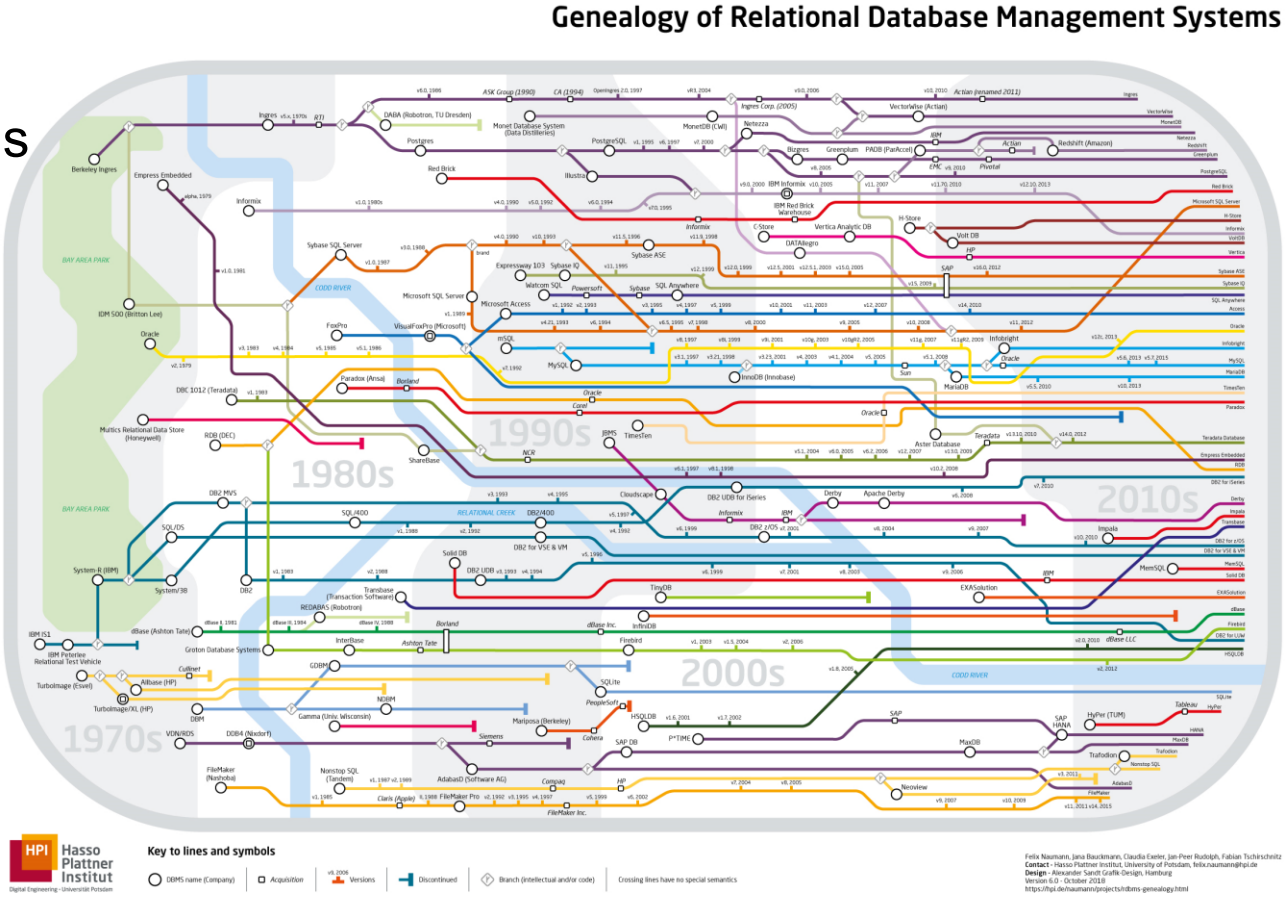
Data management – systems & platforms

- Large variety (for relational data and beyond)
- No one-size-fits-all solution, different solutions fit different needs.
- Data scientists need to know which data management solution best fits their needs.



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Schema design

- A schema defines the structure, organization, and relationships of data stored in a database.
- A clear structure of data supports data scientists in
  - Understanding the data
  - Efficient data exploration
  - Identification of data quality issues
  - Definition of data transformation
  - Model building
  - …

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Schema design example

- A relational schema structures data into tables with attributes (colums); attributes have data types, data in tuples (rows), schema defines relationships, …
- A good relational schema is normalized.

CID	City	Order
1	KL	{P3, P506, …}
504	Kuala Lumpur	{P1, P205, …}
367	SG	{P3, P5, …}
1	KL	{P2, P2, P765,…}

PoorSchemaDB

Code	City
KL	Kuala Lumpur
SG	Singapore

CID	CityCode
1	KL
504	KL
367	SG

CID	OID	PID	Amount
1	1	P3	1
1	1	P506	1
…	…	…	…
1	4	P2	2

GoodSchemaDB

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Querying data

- Different types of queries (analytical queries, search, top-k queries, …)
- Different languages with different expressive power
- Efficient and scalable query execution strategies exist for each type of query / language.
- Data scientists need to know to
  - Efficiently retrieve the data they need
  - Effectively and efficiently perform data cleaning on large volumes of data
  - Implement easy selection of data subsets for model evaluation
  - …

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Querying data example

```python
import pandas as pd

# Read employee data from CSV
df = pd.read_csv('employee_data.csv')

# Filter for employees in the 'Sales' department
sales_employees = df[df['department'] == 'Sales']

# Sort employees by salary in descending order
sorted_sales_employees = sales_employees.
              sort_values(by='salary', ascending=False)

# Get the top 5 highest-paid employees
top_5_sales_employees = sorted_sales_employees.head(5)

print(top_5_sales_employees)
```

```sql
SELECT *
FROM employees
WHERE department = 'Sales'
ORDER BY salary DESC
LIMIT 5;
```

Get top-5 of best paid sales employees (Python vs SQL)

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Data management - Wrap-up

- Knowledge about data management is crucial for data scientist to devise efficient, effective, and scalable data processing.
- The choice of a data management system, schema design, and query language can significantly impact the performance of data science pipeline development and maintenance.

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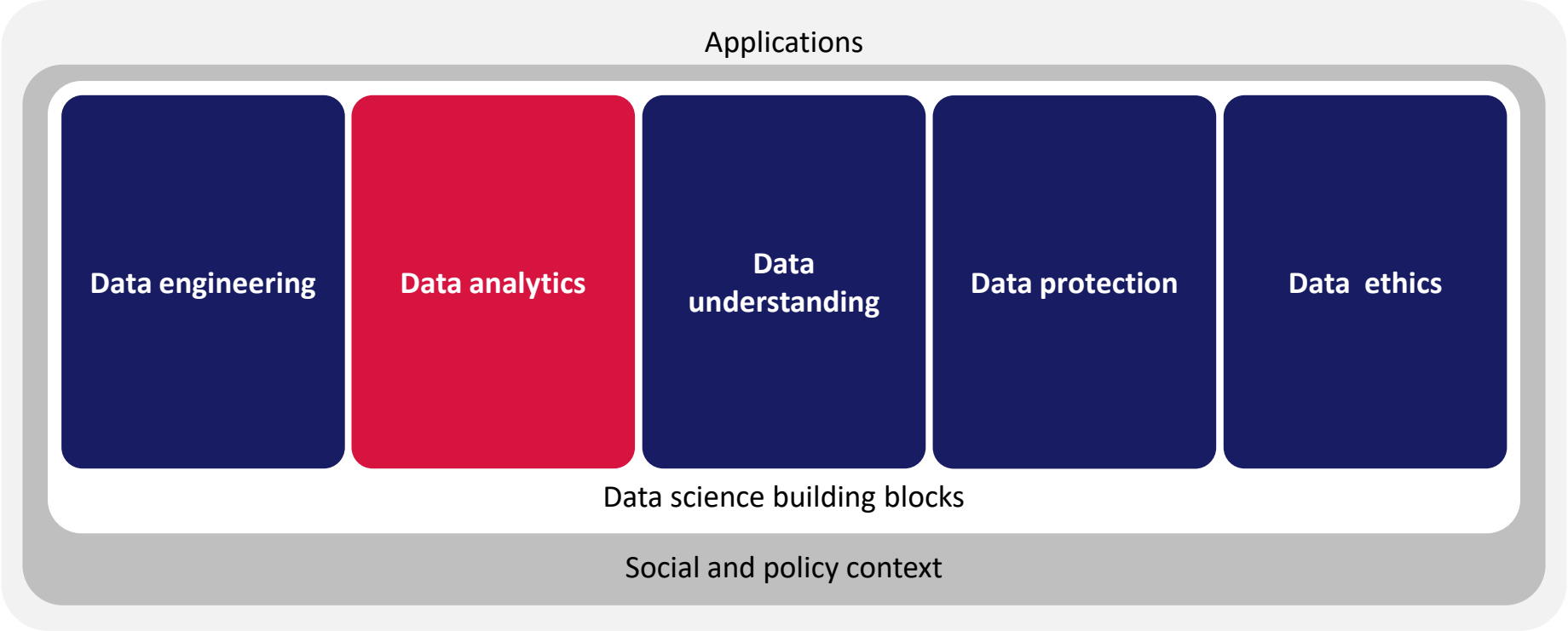
Data Science ecosystem



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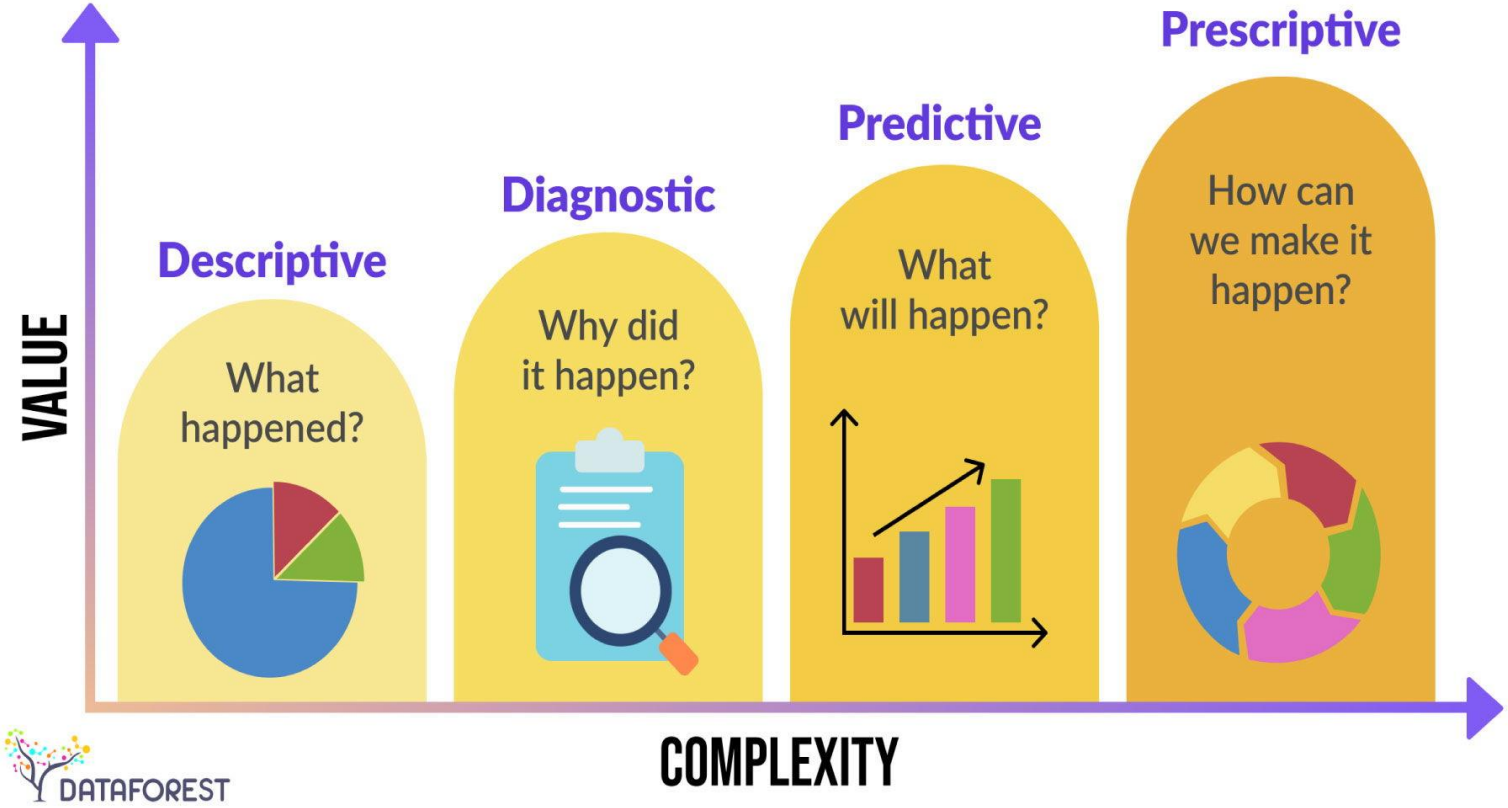
TYPES OF DATA ANALYTICS



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Descriptive Analytics

- Summarizes and describes historical data.
- Allows to analyze patterns, trends,and relationships.
- Provides insights on “What happened?” or “What does the data tell us?”.

PATIENT DEMOGRAPHICS (HEALTHCARE)

RECRUITMENT CHANNEL EFFECTIVENESS (HUMAN RESOURCES)

SALES PERFORMANCE ANALYSIS (SALES AND MARKETING)

SHIPPING AND DELIVERY TIMELINE MONITORING (SUPPLY MANAGEMENT)

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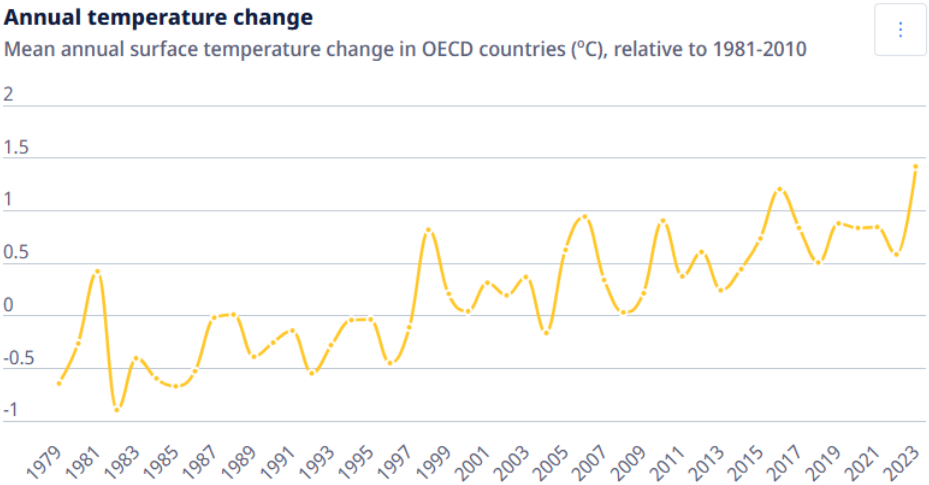

Source: IEA/OECD, "Climate-related hazards: Exposure to extreme temperature", *OECD Environment Statistics* (database).

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Diagnostic Analytics

- Studies reasons for observed trends, patterns or relationships in historical data.
- Allows to identify root causes and factors underlying observed phenomena.
- Provides insights on “Why did something happen?”.

FRAUD DETECTION (FINANCE)

DELAY ANALYSIS (LOGISTICS)

CUSTOMER CHURN ANALYSIS (SALES AND MARKETING)

WORKPLACE SAFETY INCIDENT ANALYSIS (HUMAN RESOURCES)

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Diagnostic Analytics

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- Allows to identify root causes and factors underlying observed phenomena.
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Table 1 Correlation and information flow between observed global surface temperature and different external forcing’s and internal climate variations.

From: On the causal structure between CO_2 and global temperature

Radiative Forcing	Correlation and Causality–HADCRUT4		
	Correlation	*Forcing→GMTA*[nat/year]	*GMTA→Forcing*[nat/year]
Total forcing	0.804 ± 0	0.244 ± 0.091	0.036 ± 0.080
Anthropogenic	0.863 ± 0	0.355 ± 0.112	−0.008 ± 0.005
All GHG	0.852 ± 0	0.318 ± 0.108	−0.005 ± 0.003
CO2	0.852 ± 0	0.316 ± 0.108	−0.003 ± 0.003
Aerosol	−0.810 ± 0	0.232 ± 0.095	−0.002 ± 0.006
Cloud	−0.796 ± 0	0.208 ± 0.092	−0.001 ± 0.004
Solar	0.616 ± 0	0.082 ± 0.059	0.035 ± 0.051
Volcanic	0.089 ± 0.267	0.003 ± 0.006	−0.004 ± 0.009

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Predictive Analytics

- Relies on historical data and statistical models to predict future trends our outcome.
- Supports the definition of data-based strategies or decisions affecting future outcomes.
- Provides insights on “What will happen?”.

PERSONALIZED RECOMMENDATIONS (CUSTOMER EXPERIENCE)

DEMAND FORECASTING (MANUFACTURING AND MAINTENANCE)

EARLY DISEASE PREDICTION (HEALTHCARE)

GRID OPTIMIZATION (ENERGY AND SUSTAINABILITY)

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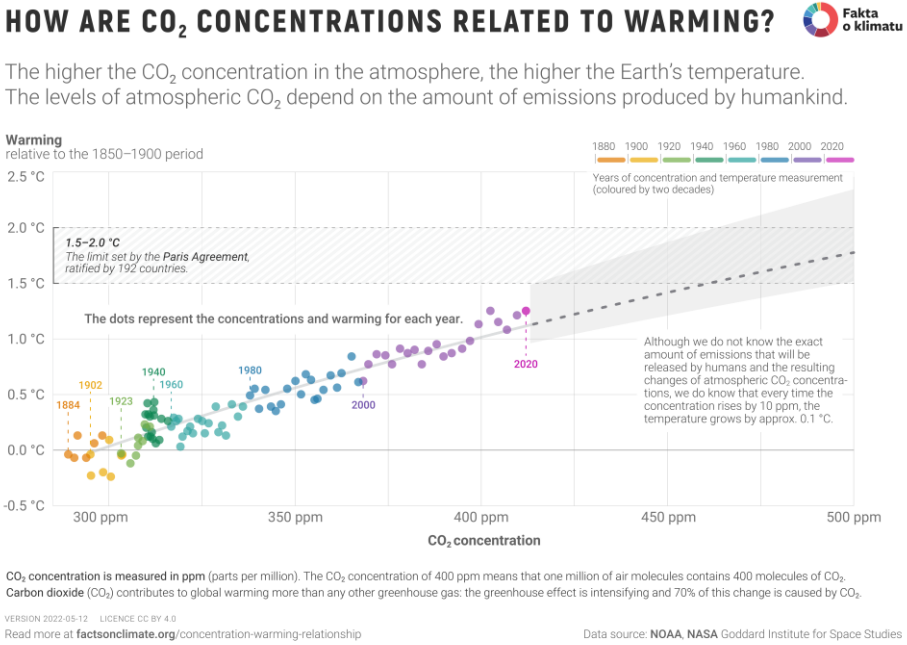

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Prescriptive Analytics

- Recommends actions towards reaching a desired result based on data.
- Supports defining measures to reach a desired outcome in the future.
- Provides insights on “What should we do?” or “How can we make it happen?”.

WORKFLOW OPTIMIZATION (FINANCE AND OPERATIONS)

SALES TERRITORY OPTIMIZATION (RETAIL AND SALES)

TREATMENT RECOMMENDATION (HEALTHCARE)

DYNAMIC PRICING STRATEGIES (TELECOM)

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Prescriptive Analytics

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Share

SITA OptiClimb®, a digital inflight prescriptive analytics tool for fuel optimization, has been selected by Singapore Airlines to support the carrier’s goal of achieving net-zero carbon emissions by 2050.

By deploying SITA OptiClimb®, the airline is able to optimize fuel utilization during the aircraft’s climb-out phase. The unique solution combines aircraft tail-specific machine-learning models with 4D weather forecasts to recommend customized climb speeds at different altitudes. It leverages historical flight data to predict fuel burn in different flight scenarios and recommends optimized climb profiles on a user-friendly interface for pilots.

It is estimated that airlines can derive fuel savings of up to 5% during climb-out on each flight, with around 5.6 million tons of carbon dioxide emissions avoided annually if every airline worldwide uses SITA OptiClimb®.

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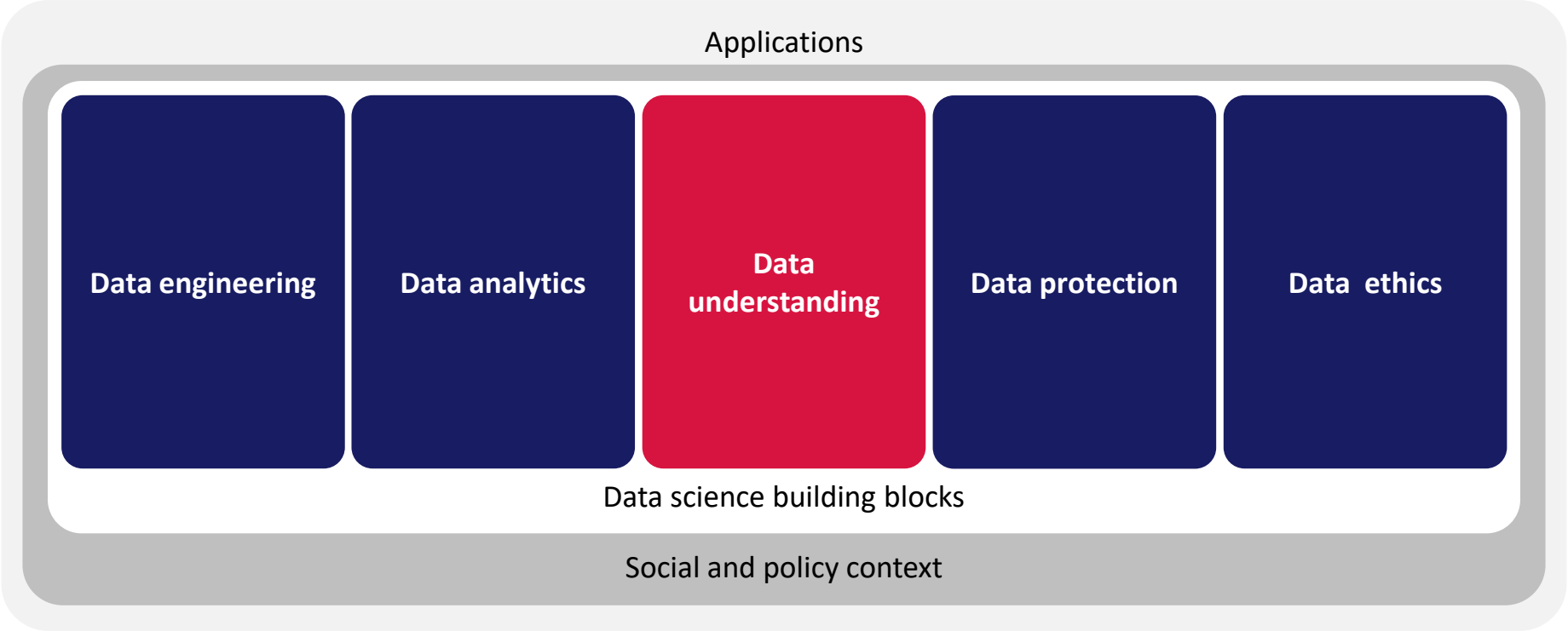
Data Science ecosystem



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Data understanding

- Data science aims at gaining **insights from data** for a particular application / goal.
- Being able to **understand** and **interpret** results is essential to gain **confidence** in them / **trust** the result and the process producing them.
- Such data understanding is **relevant in various phases of the data science process**.
- Techniques to explain or more easily understand data and data-derived results include **explanation** and **visualization**.

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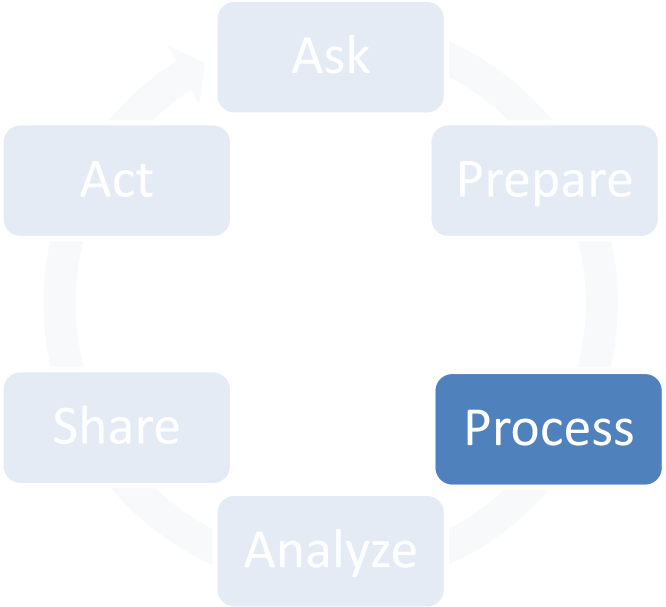
Example: Explanation during processing

Why is there a City missing?
→ location extracted from source could not be transformed to a valid US city.

Username	Name	Age	City
helloKitty	Jane Smith	46	Los Angeles
JD	John Doe	24	New York
SecretUser007	James Bond	65	⊥
maxMiller	Max Miller	37	Miami



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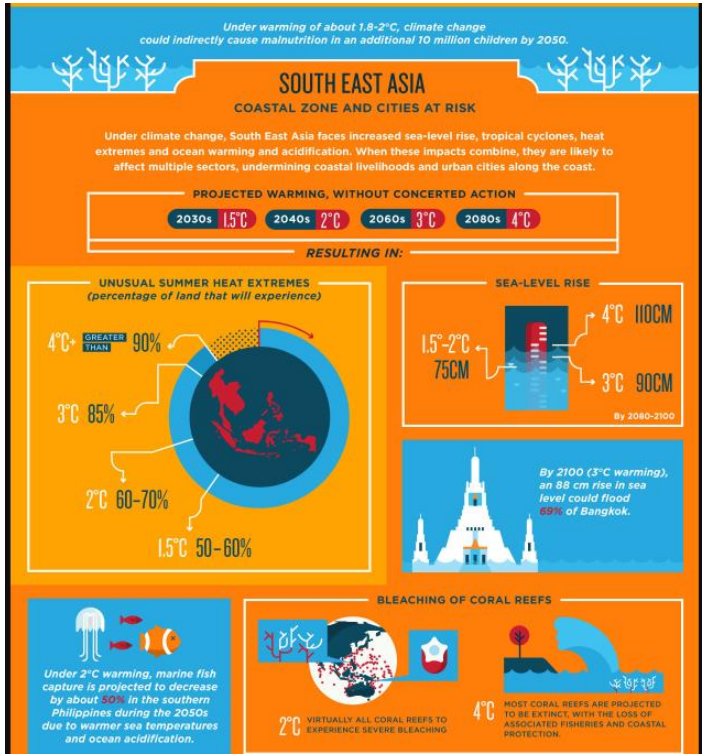
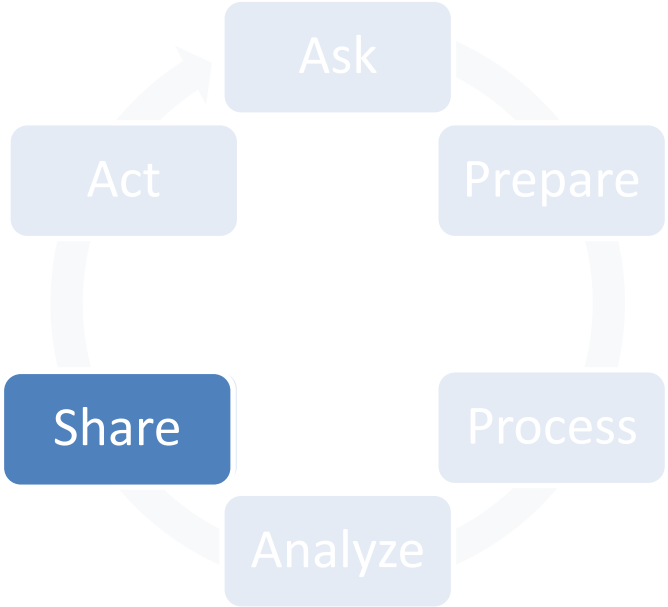
Example: Visualization to share insights





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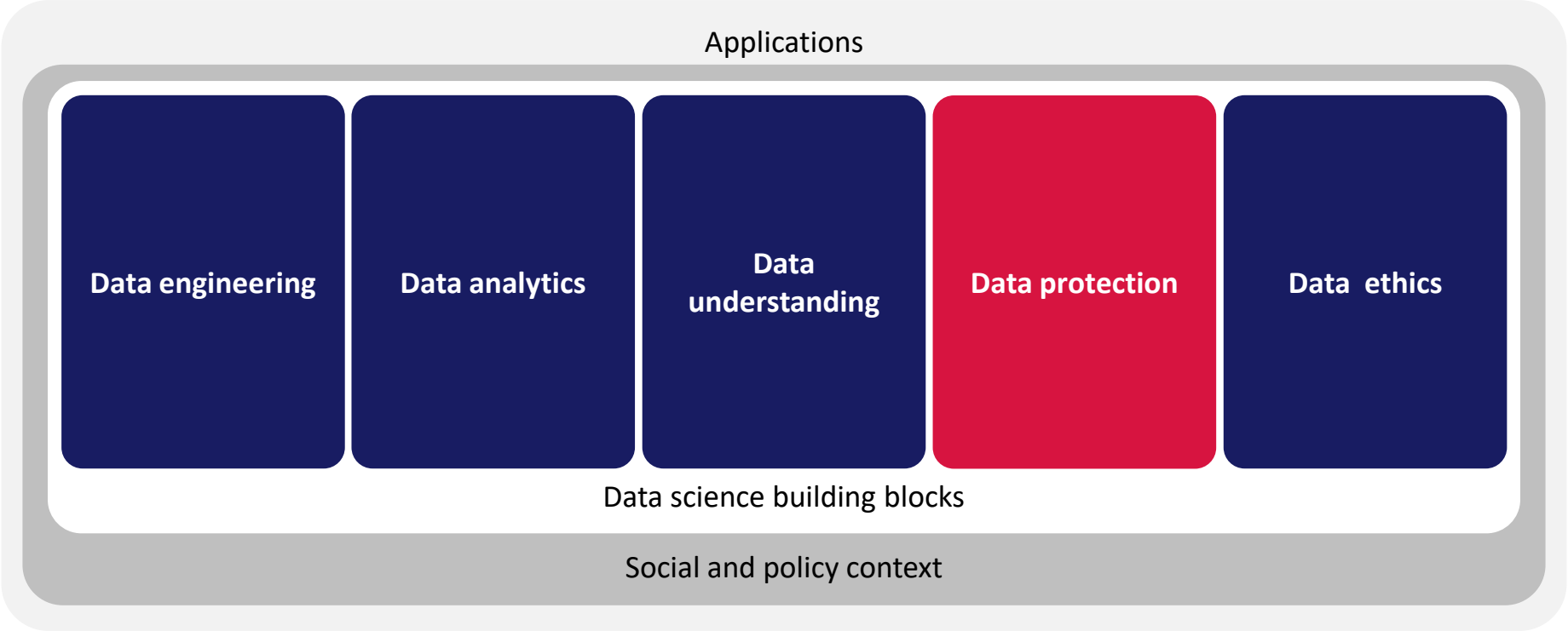
Data Science ecosystem



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Data protection

Data security

- Protects information from unauthorized access or malicious attacks.
- Deals with data confidentiality, access control, infrastructure security, and system monitoring.
- Uses technologies such as encryption, trusted execution environments, and monitoring tools.

Data privacy

- Focuses on the rights of users and groups over data about themselves
- Deals with privacy policies and regulations, management of data use by third parties, and user consent.
- Supported by privacy-enhancing technologies such as data masking, differential privacy, federated learning.

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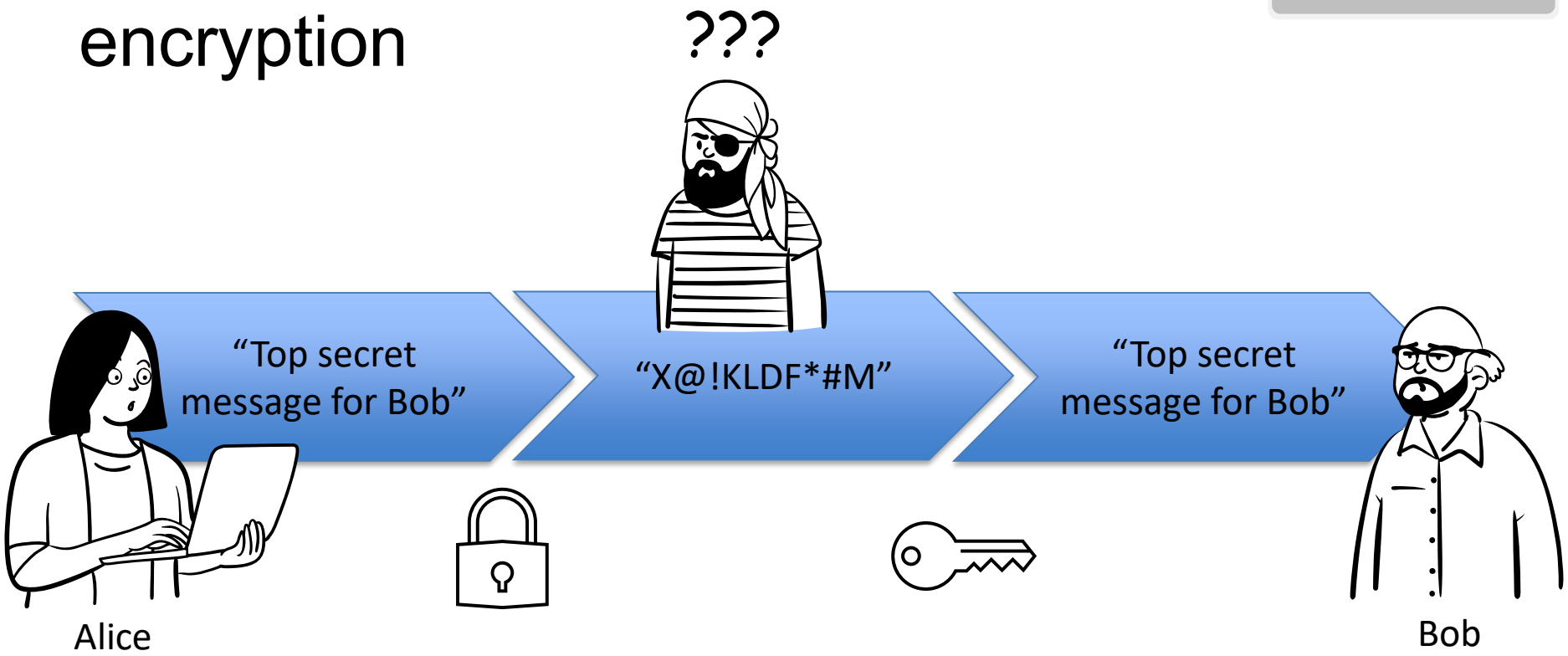
Example: Security through encryption



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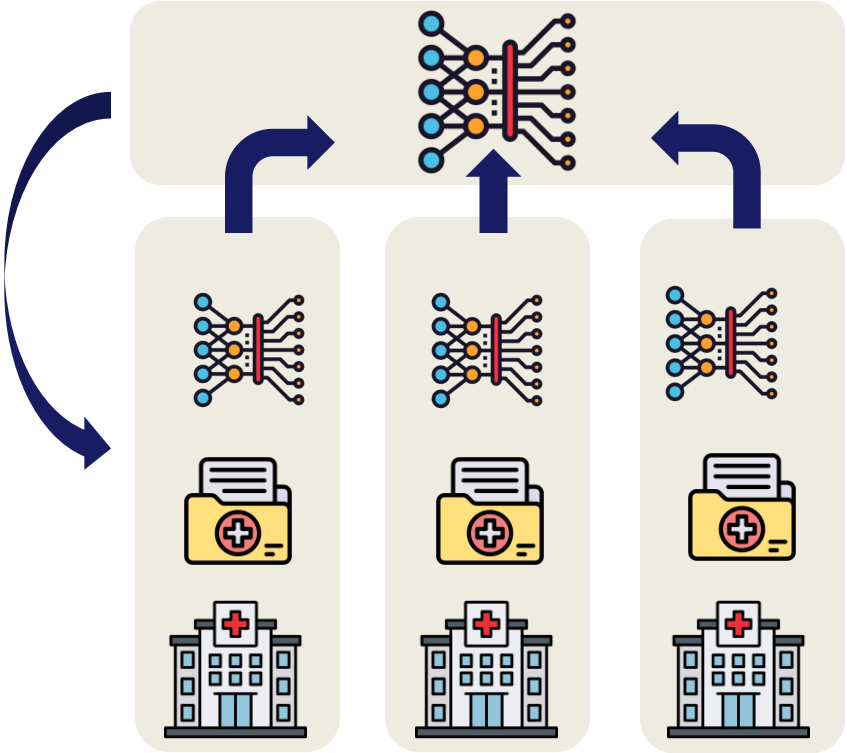
Example: Privacy through federated learning



Model back-distribution

Global model aggregation

Sharing of updated model parameters

Local training

Medical records

Hospitals

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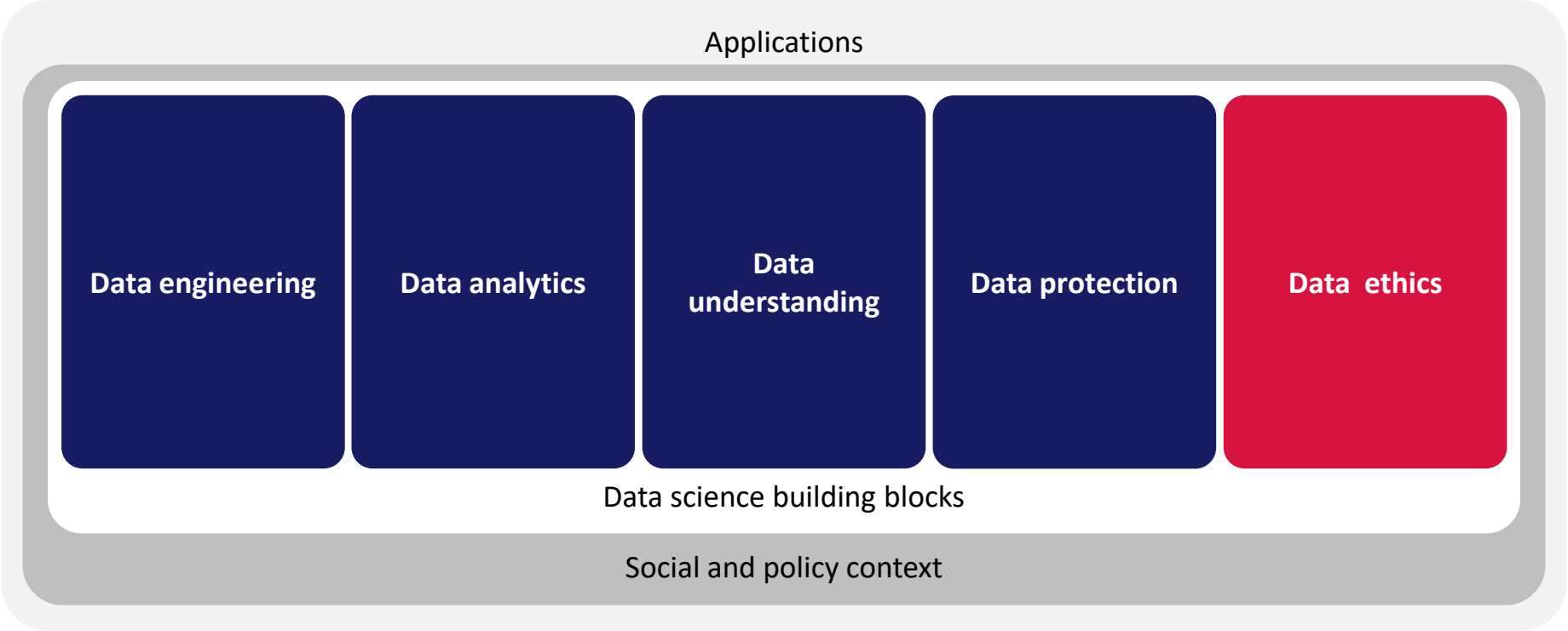
Data Science ecosystem



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Data ethics considerations

- **Individuals**, **organizations**, and **society** impacted by data-based decisions.
- Data ethics (data science ethics) considers **moral problems** related to **data**, **algorithms**, and related **practices** towards devising morally good solutions.

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Example: Ethics of data

- Data bias may lead to unfair treatment of a person or groups of persons due to over- or under-representation of persons with similar features in the training data.
- Example: face-recognition systems predominantly trained on faces of white males.

Payout for Uber Eats driver over face scan bias case

26 March 2024

Share Save

Shiona McCallum
Technology reporter

A black Uber Eats driver has received a payout after "racially discriminatory" facial-recognition checks prevented him accessing the app to secure work.

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Example: Ethics of algorithms

- Algorithmic bias caused through inadequate engineering and testing of the data analysis.
- Example: poor predictions of A-level grades in the UK.

The UK's A-level grading fiasco

For those who haven't followed the story – two weeks ago, thousands of students in England and Wales received their "A-level" exam grades. Instead of scoring actual exams, however, grades were determined by an algorithm. Almost 40% of students received grades lower than they had anticipated, sparking public outcry and legal action. Faced with protests, the UK government retracted the grades. Students will now receive grades based on their teacher's estimate of what their grade would have been, had the exams gone forward as planned.

How did the algorithm discriminate?

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Example: Ethics of practices

- Responsibility, liability, ethical behavior of organizations in charge of data processes, strategies, and policies.
- Example: see previous slides + regulators enforcing ethical practice

IntelliVision censured for misleading biometric accuracy and bias claims by FTC

Claims undermined by NIST test results

Dec 3, 2024, 12:29 pm EST | Chris Burt

CATEGORIES Biometrics News | Facial Recognition | Trade Notes

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Data Science Ecosystem

SUMMARY

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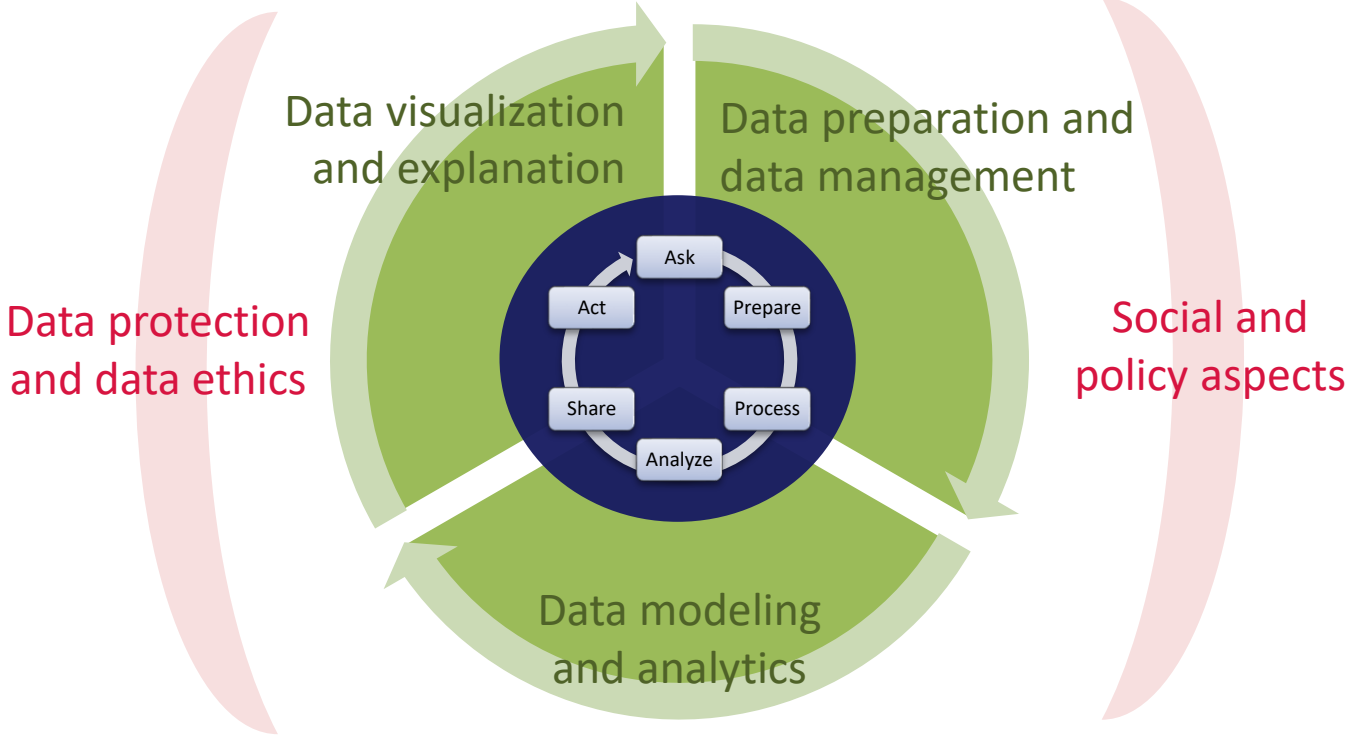
Applications, systems, and deployments

Data Science application

Data Science system

Data Science application deployment



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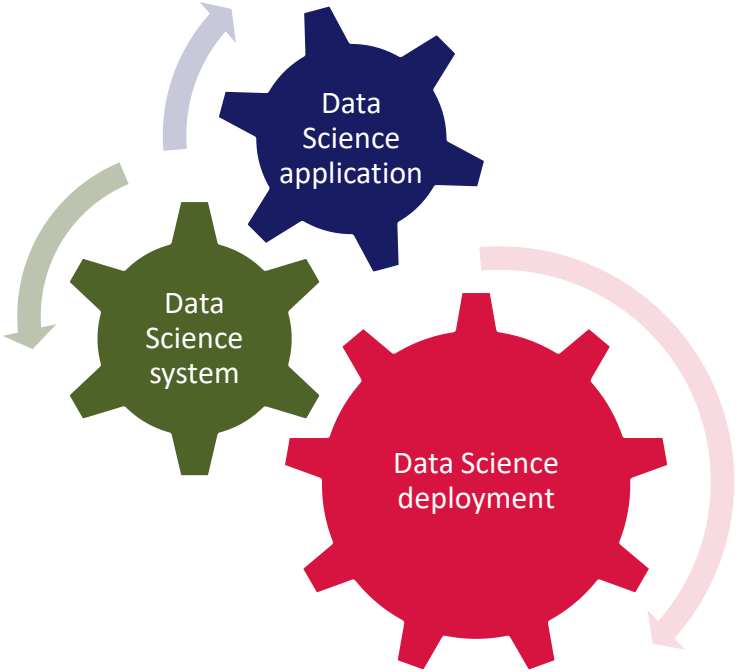
Holistic consideration



- Applications, systems, and deployments are interlinked and influence one another.
- Examples:
  - Data science systems provide tools and technologies that help data scientists address their problems.
  - Data science systems need to be carefully designed to fit the target applications.
  - Data science systems need to support deployments within their social and policy context.

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Data Science ecosystem

Applications

Data engineering

- Data preparation
- Data management

Data analytics

- Data exploration (data mining)
- Models and algorithms (machine learning)

Data understanding

- Explanation
- Visualization

Data protection

- Security
- Privacy

Data ethics

- Impact on individuals, organizations, and society
- Bias in data
- Algorithmic bias
- Regulatory issues

Data science building blocks

Social and policy context

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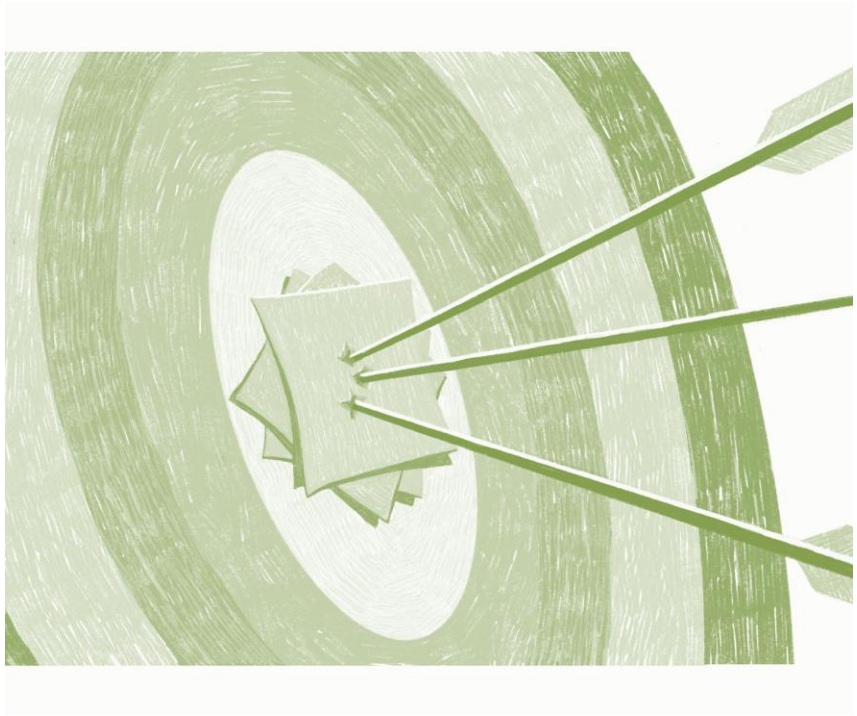
You should now be able to…

- … describe the main components of the data science ecosystem.
- … explain how different factors influence the development of a data science system.
- … define the functionality of data science building blocks at a high level to solve data science applications.
- … reflect on possible data protection, data ethics, social and policy aspects that may affect the data science application deployment.



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